

Standalone numerical study for the calibrated gDMR parameter set with positive rate

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1 Numerical studies

This standalone document repeats the manuscript numerical-study design for the calibrated gDMR parameter set reported in Table 1, using the positive risk-free rate $r = 0.02$. The option is a Bermudan put with twelve exercise dates, maturity $T = 1$, and strikes $K = 70, 80, 90, 100, 110$. The direct plain LSMC estimator is compared with the Hybrid LSMC–PDE estimator. All prices, standard errors, confidence intervals, and relative errors are computed from the direct estimators, and benchmark-relative errors use the plain LSMC references in Table 2.

The benchmark references are obtained with 1200 Euler time steps and 1,200,000 direct paths. Across the five strikes, the benchmark direct prices range from 2.522851 to 15.743520. These values define the reference scale used in the step-sweep and path-sweep experiments below.

1.1 Fixed path budget, varying number of Euler time steps

The first experiment keeps the direct path budget fixed and varies the number of Euler time steps. The two path budgets are 20,000 and 60,000, and both methods are evaluated at 24, 48, 72, and 96 steps. This isolates how the time discretization affects the direct Bermudan estimator away from the benchmark regime.

For 20,000 paths, the LSMC relative-error range over the full step-sweep grid is 0.344%–8.645%, while the Hybrid LSMC–PDE range is 0.019%–7.459%. Figure 1 shows the strike-wise relative-error profiles, and Table 3 reports the representative 72-step values.

For 60,000 paths, the same design gives an LSMC relative-error range of 0.102%–7.720% and a Hybrid LSMC–PDE range of 0.028%–7.857%. Figure 2 and Table 4 give the corresponding graphical and tabular summaries.

Table 1 Model and numerical parameters for the standalone calibrated positive-rate study.

Quantity	Value
Model	gDMR
Option	Bermudan put
S_0	100.0
K	70, 80, 90, 100, 110
T	1.0
r	0.02
v_0	0.114
v'_0	0.110
κ_1	5.5
κ_2	0.1
θ	0.078
ξ_1	2.689
ξ_2	0.502
ρ_{12}	-0.982
ρ_{13}	-0.727
ρ_{23}	0.590
δ_1	0.94
δ_2	0.94
Exercise dates	12
Benchmark	LSMC, 1200 steps, 1,200,000 paths
Hybrid asset grid	301

Table 2 Benchmark direct LSMC reference prices for the standalone study.

Strike	Price	SE	CI lower	CI upper
70	2.522851	0.007103	2.508929	2.536773
80	4.298933	0.009345	4.280617	4.317249
90	6.942119	0.011774	6.919042	6.965196
100	10.682037	0.014105	10.654391	10.709683
110	15.743520	0.015757	15.712636	15.774404

1.2 Fixed number of Euler time steps, varying path counts

The second experiment fixes the number of Euler time steps and varies the direct path budget. The reported path grid is 250, 1000, 5000, 10000, 20000, 40000, and 60000. The two discretization levels are 48 and 60 steps.

At 48 steps for the at-the-money strike, the LSMC relative-error range over all path counts is 1.105%–25.948%, while the Hybrid LSMC–PDE range is 0.088%–11.118%. Figure 3 reports the full strike-wise path sweep, and Table 5 reports the 20,000-path representative values.

At 60 steps for the at-the-money strike, the corresponding LSMC range is 1.324%–28.932%, and the Hybrid LSMC–PDE range is 0.049%–10.102%. Figure 4 and Table 6 provide the companion summaries.

The two experiment families provide a direct check of the pricing behavior under the calibrated positive-rate parameter set. The tables should be read together with the confidence intervals, since both methods retain finite Monte Carlo variability at

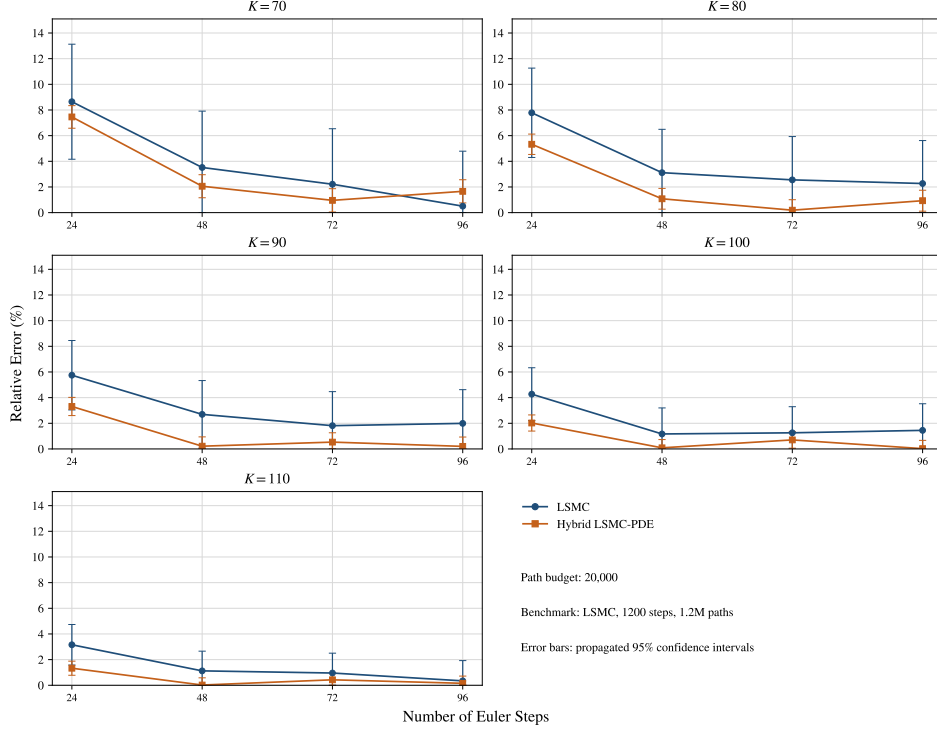


Fig. 1 Relative errors for the matched 20,000-path step-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Table 3 Representative direct metrics at 20,000 paths and 72 Euler steps.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	2.578706	2.214%	2.547003	0.957%
80	4.408717	2.554%	4.307120	0.190%
90	7.068072	1.814%	6.905102	0.533%
100	10.816495	1.259%	10.606747	0.705%
110	15.893853	0.955%	15.676443	0.426%

Table 4 Representative direct metrics at 60,000 paths and 72 Euler steps.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	2.578570	2.209%	2.556746	1.344%
80	4.361900	1.465%	4.326225	0.635%
90	7.032832	1.307%	6.940184	0.028%
100	10.793695	1.045%	10.662148	0.186%
110	15.853083	0.696%	15.749100	0.035%

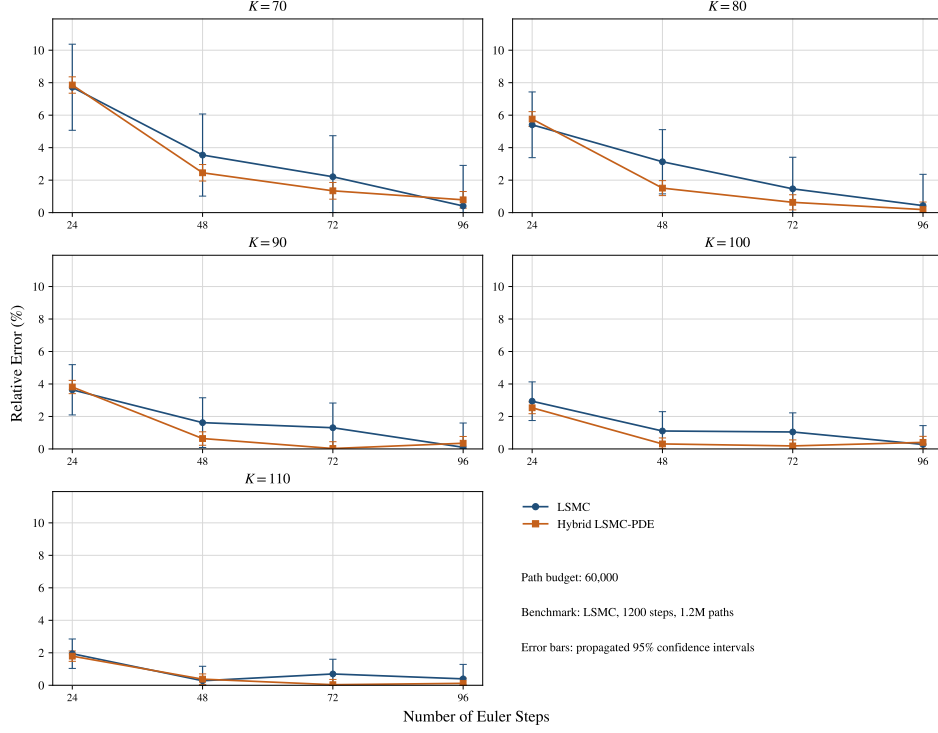


Fig. 2 Relative errors for the matched 60,000-path step-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Table 5 Representative direct metrics at 20,000 paths under the fixed 48-step path sweep.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	2.611646	3.520%	2.574765	2.058%
80	4.432813	3.114%	4.345363	1.080%
90	7.129239	2.695%	6.957089	0.216%
100	10.806834	1.168%	10.672650	0.088%
110	15.920337	1.123%	15.746530	0.019%

Table 6 Representative direct metrics at 20,000 paths under the fixed 60-step path sweep.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	2.635093	4.449%	2.563087	1.595%
80	4.483550	4.294%	4.317574	0.434%
90	7.119832	2.560%	6.910022	0.462%
100	10.909701	2.131%	10.609574	0.678%
110	15.917591	1.106%	15.679793	0.405%

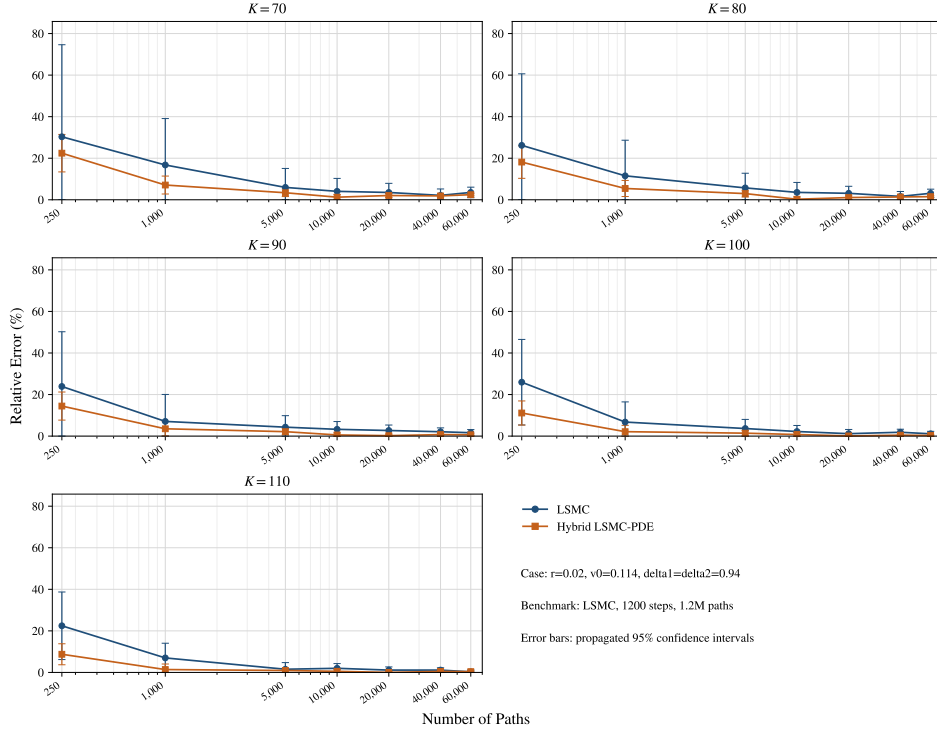


Fig. 3 Relative errors for the fixed 48-step path-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

the moderate path budgets used in the sweeps. The appendix-style tables below list the underlying prices used to form the figures and representative summaries.

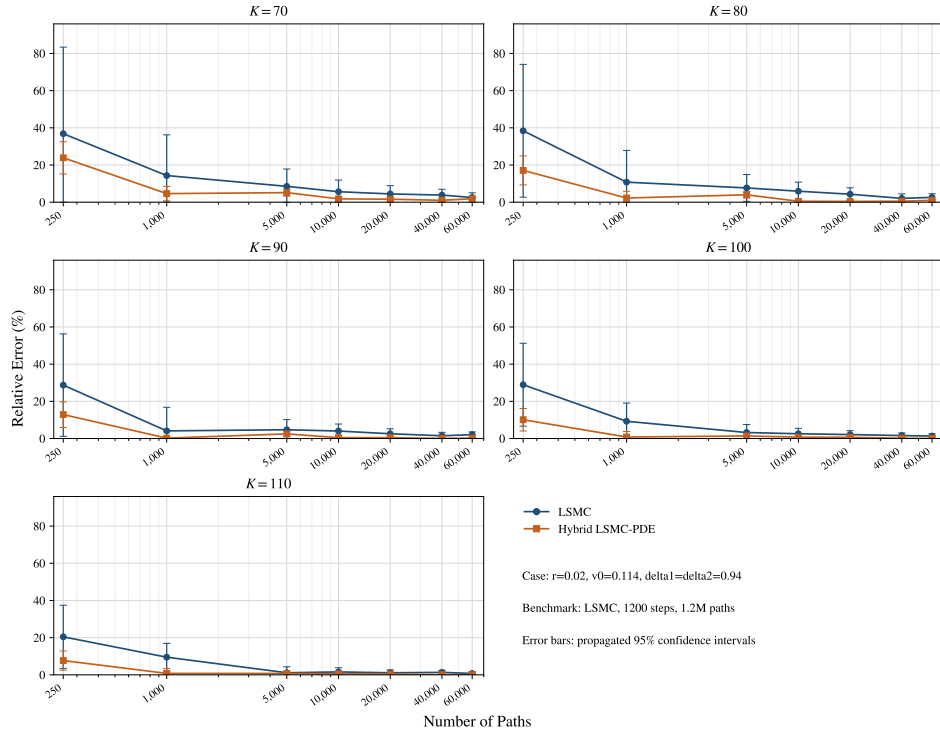


Fig. 4 Relative errors for the fixed 60-step path-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Positive-rate gDMR price tables

Table 7 Benchmark prices. Prices are rounded to three decimals.

Strike	70	80	90	100	110
LSMC benchmark	2.523	4.299	6.942	10.682	15.744

Table 8 Fixed-path step-sweep prices with 20,000 paths. Prices are rounded to three decimals.

Method/step	24	48	72	96
LSMC K=70	2.741	2.612	2.579	2.536
Hybrid LSMC-PDE K=70	2.711	2.575	2.547	2.565
LSMC K=80	4.634	4.433	4.409	4.396
Hybrid LSMC-PDE K=80	4.528	4.345	4.307	4.339
LSMC K=90	7.341	7.129	7.068	7.080
Hybrid LSMC-PDE K=90	7.172	6.957	6.905	6.956
LSMC K=100	11.138	10.807	10.816	10.837
Hybrid LSMC-PDE K=100	10.898	10.673	10.607	10.679
LSMC K=110	16.241	15.920	15.894	15.798
Hybrid LSMC-PDE K=110	15.953	15.747	15.676	15.768

Table 9 Fixed-path step-sweep prices with 60,000 paths. Prices are rounded to three decimals.

Method/step	24	48	72	96
LSMC K=70	2.718	2.612	2.579	2.533
Hybrid LSMC-PDE K=70	2.721	2.585	2.557	2.543
LSMC K=80	4.531	4.434	4.362	4.318
Hybrid LSMC-PDE K=80	4.547	4.364	4.326	4.307
LSMC K=90	7.195	7.055	7.033	6.949
Hybrid LSMC-PDE K=90	7.207	6.987	6.940	6.918
LSMC K=100	10.996	10.800	10.794	10.712
Hybrid LSMC-PDE K=100	10.953	10.715	10.662	10.639
LSMC K=110	16.050	15.788	15.853	15.806
Hybrid LSMC-PDE K=110	16.026	15.803	15.749	15.725

Table 10 Fixed 48-step path-sweep prices. Prices are rounded to three decimals.

Method/strike	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC K=70	3.288	2.946	2.673	2.625	2.612	2.576	2.612
Hybrid LSMC-PDE K=70	3.089	2.702	2.608	2.554	2.575	2.570	2.585
LSMC K=80	5.425	4.795	4.545	4.452	4.433	4.369	4.434
Hybrid LSMC-PDE K=80	5.079	4.533	4.427	4.310	4.345	4.357	4.364
LSMC K=90	8.601	7.433	7.241	7.169	7.129	7.088	7.055
Hybrid LSMC-PDE K=90	7.946	7.187	7.090	6.903	6.957	6.992	6.987
LSMC K=100	13.454	11.404	11.072	10.918	10.807	10.881	10.800
Hybrid LSMC-PDE K=100	11.870	10.910	10.833	10.601	10.673	10.730	10.715
LSMC K=110	19.278	16.843	15.985	16.057	15.920	15.921	15.788
Hybrid LSMC-PDE K=110	17.120	15.962	15.889	15.660	15.747	15.819	15.803

Table 11 Fixed 60-step path-sweep prices. Prices are rounded to three decimals.

Method/strike	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC K=70	3.452	2.884	2.738	2.665	2.635	2.619	2.587
Hybrid LSMC-PDE K=70	3.125	2.639	2.653	2.568	2.563	2.549	2.570
LSMC K=80	5.950	4.762	4.629	4.553	4.484	4.387	4.411
Hybrid LSMC-PDE K=80	5.034	4.396	4.470	4.321	4.318	4.321	4.342
LSMC K=90	8.936	7.227	7.270	7.222	7.120	7.045	7.084
Hybrid LSMC-PDE K=90	7.836	6.957	7.115	6.910	6.910	6.939	6.957
LSMC K=100	13.773	11.673	11.026	10.955	10.910	10.850	10.823
Hybrid LSMC-PDE K=100	11.761	10.596	10.831	10.604	10.610	10.666	10.677
LSMC K=110	18.962	17.240	15.920	15.991	15.918	15.952	15.860
Hybrid LSMC-PDE K=110	16.953	15.621	15.863	15.659	15.680	15.755	15.757